

Polynomial Regression Modeling of EEG and Audio Signals

Task 2.3 Final Report

1 Introduction

This study investigates the neural modulation of two distinct brain regions during guided meditation. The dataset comprises electroencephalography (EEG) measurements collected during a neuroscience experiment where participants listened to a meditation guide. The specific areas monitored were the right auditory cortex, which is dedicated to processing auditory sensations, and the prefrontal cortex, which governs executive function, planning, and consciousness. The primary objective is to mathematically model the relationship between the meditation guide's voice (the audio signal) and the corresponding brain activity in these two regions.

2 Literature Review & Hypothesis

Previous understandings of brain functionality suggest different processing complexities depending on the region. Based on this, the researchers hypothesized that the auditory cortex (x_2) is linearly related to the audio signal (y), given its direct role in sensory processing. In contrast, a nonlinear relationship is predicted for the prefrontal cortex (x_1), reflecting its involvement in complex, higher-order cognitive tasks. This project tests this hypothesis by evaluating a candidate set of nonlinear polynomial regression models to identify the 'true' mathematical relationship.

3 Methodology

To determine the most suitable mathematical model, we evaluated five specific candidate polynomial regression models that combine linear and nonlinear terms of the prefrontal cortex signal (x_1) and the auditory cortex signal (x_2). The candidate models are structured as follows:

- **Model 1:** $y = \theta_1 x_1^3 + \theta_2 x_2^5 + \theta_{bias} + \epsilon$
- **Model 2:** $y = \theta_1 x_1^4 + \theta_2 x_2^2 + \theta_{bias} + \epsilon$
- **Model 3:** $y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias} + \epsilon$
- **Model 4:** $y = \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_2^3 + \theta_{bias} + \epsilon$
- **Model 5:** $y = \theta_1 x_1^3 + \theta_2 x_1^4 + \theta_3 x_2 + \theta_{bias} + \epsilon$

The parameters (θ) for each model were estimated using the Least Squares matrix method:

$$\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (1)$$

The models were subsequently evaluated using the following statistical criteria:

1. **Residual Sum of Squares (RSS):** To measure the variance of the model's prediction errors.

$$RSS = \sum_{i=1}^n (y_i - \mathbf{x}_i \hat{\theta})^2 \quad (2)$$

2. **Log-likelihood function:** To assess goodness of fit, assuming the additive noise is independent and identically distributed (i.i.d) Gaussian.

$$\ln p(D|\hat{\theta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2} RSS \quad (3)$$

3. **Akaike Information Criterion (AIC) & Bayesian Information Criterion (BIC):** To identify the optimal model by balancing goodness-of-fit against model complexity (penalizing models with a higher number of parameters, k).

$$AIC = 2k - 2 \ln p(D|\hat{\theta}) \quad (4)$$

$$BIC = k \cdot \ln(n) - 2 \ln p(D|\hat{\theta}) \quad (5)$$

4. **Q-Q Plots:** To visually verify that the model prediction errors (residuals) closely follow a Normal/Gaussian distribution.

4 Python Code Implementation

The following code was executed to conduct the preliminary data analysis and evaluate the candidate regression models.

```

1 import numpy as np
2 import pandas as pd
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import statsmodels.api as sm
6 import scipy.stats as stats
7
8 # Load the dataset
9 url = 'https://raw.githubusercontent.com/sijuswamy/A-modern-approach-to-
    -Research/main/X.csv'
10 df = pd.read_csv(url)
11
12 # Assign variables
13 y = df['y'].values
14 x1 = df['x1'].values
15 x2 = df['x2'].values
16 n = len(df)
17
18 # Define Design Matrices for all 5 candidate models
19 X1 = np.column_stack((x1**3, x2**5, np.ones(n)))
20 X2 = np.column_stack((x1**4, x2**2, np.ones(n)))

```

```

21 X3 = np.column_stack((x1**3, x2, x1, np.ones(n)))
22 X4 = np.column_stack((x1, x1**2, x1**3, x2**3, np.ones(n)))
23 X5 = np.column_stack((x1**3, x1**4, x2, np.ones(n)))
24
25 models = {'Model 1': X1, 'Model 2': X2, 'Model 3': X3, 'Model 4': X4, '
    Model 5': X5}
26 model_metrics = []
27
28 # Compute metrics for each model
29 for name, X in models.items():
30     k = X.shape[1] # Number of parameters
31
32     # Direct OLS Matrix Method
33     theta_hat = np.linalg.inv(X.T @ X) @ X.T @ y
34
35     # Model Residual Sum of Squared Errors (RSS)
36     y_pred = X @ theta_hat
37     residuals = y - y_pred
38     RSS = np.sum(residuals**2)
39
40     # Log-likelihood, AIC, and BIC
41     sigma_sq_hat = RSS / n
42     log_likelihood = -(n/2)*np.log(2*np.pi) - (n/2)*np.log(sigma_sq_hat)
43     - (1/(2*sigma_sq_hat))*RSS
44     AIC = 2*k - 2*log_likelihood
45     BIC = k*np.log(n) - 2*log_likelihood
46
47     model_metrics.append({'Model': name, 'RSS': RSS, 'AIC': AIC, 'BIC':
48     BIC})
49
50 # Display final results
51 results_df = pd.DataFrame(model_metrics)
52 print(results_df)

```

5 Summary of Findings

The preliminary exploratory data analysis revealed distinct interaction patterns. The correlation matrix indicated that the prefrontal cortex EEG (x_1) shares a strong positive correlation with the audio signal (y) at approximately 0.86, whereas the auditory cortex EEG (x_2) exhibits a weak negative correlation of -0.25.

Upon evaluating the five candidate regression models, the calculated metrics clearly differentiated the models' performance:

- **Model 1:** AIC = 12936.99 | BIC = 12954.34
- **Model 2:** AIC = 19314.36 | BIC = 19331.71
- **Model 3:** AIC = 5731.54 | BIC = 5754.67
- **Model 4:** AIC = 9695.17 | BIC = 9724.08
- **Model 5:** AIC = 11536.90 | BIC = 11560.04

Model 3 achieved the lowest RSS (1525.62) and substantially lower AIC and BIC values compared to all other candidates. Furthermore, the Q-Q plots indicated that

the residuals for Model 3 most closely aligned with a normal distribution, satisfying the project’s assumption of additive Gaussian noise.

6 Conclusions

Based on the quantitative analysis, Model 3 ($y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias} + \epsilon$) is unequivocally the best regression model to describe the evoked brain responses. It provides the most optimal balance of accuracy and simplicity, as evidenced by possessing the lowest AIC and BIC scores.

Crucially, the selection of Model 3 directly validates the researchers’ initial hypothesis. In the structure of Model 3, the auditory cortex variable (x_2) appears exclusively as a linear term ($\theta_2 x_2$). Conversely, the prefrontal cortex variable (x_1) is modeled as a nonlinear polynomial encompassing a cubic term ($\theta_1 x_1^3$) alongside a linear term. Consequently, this mathematical modeling successfully proves that the auditory cortex responds linearly to the meditation guide’s voice, while the prefrontal cortex engages in a more complex, nonlinear relationship, accurately reflecting its role in higher-order executive functions.